KUNAL DAS

Full Stack Developer | Python & Django Developer
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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on expertise in Python, Django, REST APIs, and relational databases, complemented by practical experience in IoT and embedded systems. Built and deployed a production-grade multi-tenant SaaS platform with role-based access control, automated workflows, and cloud infrastructure on Render. Proven team leader who delivers complete software solutions from initial database design through production release.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (Basic), SQL, HTML, CSS

Backend Development: Django, REST APIs, API Integration, JWT Authentication, Multi-Tenancy, SaaS Architecture

Databases: PostgreSQL, MySQL — schema design

Frontend Development: HTML5, CSS3, JavaScript (Basic), Responsive Web Design

IoT and Embedded Systems: Arduino, ESP32, Ultrasonic, IR, GPS and GSM Sensors, Serial Communication, PCB Prototyping

Tools and Platforms: Git, GitHub, Render, MS Office

Core Concepts: OOP, Real-Time Systems, Database Management, Circuit Design, Agile Development

Soft Skills: Problem Solving, Project Management, Clear Communication

PROFESSIONAL EXPERIENCE

Full Stack Developer | Dooars Granthika - SaaS Library Management System *Jan 2025 - Present*

Tech Stack: Python, Django, PostgreSQL, MySQL, REST APIs, ReportLab, WhatsApp Business API

- Built a production-ready multi-tenant SaaS Library Management System in Django, implementing tenant isolation, role-based access control (RBAC), and modular architecture for long-term scalability.
- Engineered automated fine calculation, member blocking, and background notification pipelines via email and WhatsApp Business API, eliminating all manual administrative intervention.
- Designed a secure ID generation system using CSPRNG and sequential identifiers; developed analytics dashboards with PDF and CSV report export via ReportLab.
- Deployed the platform on Render with PostgreSQL, handling schema migrations, environment configuration, and membership certificate generation.

ACADEMIC PROJECTS

Team Leader | Backend and Database Developer | Library Management System (LMS) *Jan 2025 - Jun 2025*

- Developed a full-featured Library Management System in Python and MySQL, designing a normalized relational schema for books, members, and transactions.
- Implemented CRUD operations, keyword search, and automated workflow logic; managed team collaboration and version control using Git and GitHub.

Team Leader | Embedded Systems Developer | Smart Accident Prevention, Detection and Reporting System (SAPDRS) *Jul 2024 - Dec 2024*

- Designed and programmed a real-time IoT accident detection system on ESP32, integrating ultrasonic, IR, GPS, and GSM modules with C-based sensor-processing logic.
- Automated emergency SMS alerts with live GPS coordinates and validated full circuit schematics through iterative PCB prototyping and debugging.

EDUCATION

Diploma in Electronics and Telecommunication Engineering *2022 - 2025*
Falakata Polytechnic, Government of West Bengal | **GPA: 9.0 | 88.5%**

Bachelor of Science (Physics, Chemistry, Mathematics) *2018 - 2021*
University of North Bengal | **GPA: 7.18**

CERTIFICATIONS AND TRAINING

- Employability Skills – JobReady Certificate | Wadhwani Foundation
- AI for Beginners Certificate | HP LIFE
- Generative AI for Educators Certificate | Google and MIT RAISE
- Lean Startup Methodology - Certified | NPTEL+ Workshop

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.